ATOMiK — Product & Technical Overview

ATOMiK · State-Aware Compute Architecture · Pre-Seed · May 2026

DATA ROOM — TECHNICAL SUMMARY

Architecture Overview

ATOMiK is a state-aware compute primitive implemented as FPGA-validated hardware IP for specific proof artifacts, with an ASIC feasibility path still to be reviewed. The core equation: state = reference_state XOR accumulated_delta

Plain-English diagram: traditional path = full state -> scan/copy/sync/rebuild -> full state again. ATOMiK path = reference state + accumulated deltas -> coalesce where possible -> READ only when needed -> SWAP boundary.

Operations:

LOAD(addr, initial_state)	Write reference state to address slot
ACCUM(delta)	XOR a delta into the current accumulator
READ()	Reconstruct current state: reference XOR accumulated deltas
SWAP(addr)	Commit clean epoch boundary

Hardware Implementation

Platform	Zynq XC7Z020-2CLG484I (HamGeek RK-ZYNQ7020-F)
Soft CPU	VexRiscv SMP rv32ima in frozen Linux userspace baseline; NaxRiscv RV64GC SD-boot work remains build/bring-up context until promoted by run artifacts
ATOMiK core	Single-bank, 256×64-bit state table, LiteX Migen CSR
CSR base	0xf0000000 in frozen VexRiscv baseline; verify generated CSR map for each newer NaxRiscv/SD-boot build
Linux	Buildroot/Linux 6.9 in frozen userspace validation; Ubuntu/NaxRiscv work remains bring-up context unless linked to run artifacts
Bitstream	hamgeek_rk7020f.bit in frozen baseline; newer SD-boot bitstream should be cited by exact build artifact
FPGA utilization	Build-specific; do not quote utilization publicly without the matching synthesis artifact

Validated Proof

Each proof item below is intentionally separated to avoid treating different evidence labels as one blended claim.

Linux Userspace Algebraic Checks

Evidence label: HARDWARE_VALIDATED. Current summary: 16/16 algebraic property checks passed on XC7Z020 through Linux userspace.

ATOMiK Desk v0.40-A UI

Evidence label: HARDWARE_VALIDATED. Current summary: framebuffer-native prototype on live hardware.

AX7020 Board-Run Matrix

Evidence label: LIVE_MEASURED. Current summary: four-way comparison with raw artifacts and workload-specific caveats.

Formal Proof Foundation

Evidence label: FORMAL_PROOF where directly audited; otherwise SOFTWARE_VALIDATED / proof work present. Do not publish theorem-count hype until reconciled.

SD Boot Artifacts

Evidence label: BUILD_ARTIFACT. Current summary: BOOT.bin and bitstream exist; standalone power-on artifact remains gated until reproducible logs exist.

MMIO Interface

In the frozen Linux userspace validation, ATOMiK was accessible via /dev/mem at CSR base 0xf0000000. Ordering requirement: after CSR writes, issue fence iorw,iorw plus a dummy STATUS read before reading STATE_LO/HI to flush the Wishbone pipeline. Verify the generated CSR map for newer builds.

ASIC Path

The architecture is intended to be ASIC-portable, but ASIC feasibility remains a funded diligence task. The 256x64-bit state table in the frozen baseline is 16,384 bits, or 2KB of raw state storage; production area, power, memory macro, verification, and integration economics require expert review.

Pre-seed allocates $300K for ASIC feasibility (mentor-reviewed go/no-go). Tape-out is NOT in this round.

Where ATOMiK Is Less Likely To Fit

- The workload is not state-heavy.
- Every region changes uniformly or the full state truly must move every time.
- Hardware access overhead dominates the measured path.
- The current software path is already optimal for the metric that matters.
- Correctness cannot be cleanly validated.
- The customer cannot provide a trace, baseline, or success metric.

Diligence Questions We Expect

- What is the exact board, build, bitstream, and commit for each artifact?
- What is frozen validation versus current SD-boot/NaxRiscv bring-up?
- Which proof is live measured, hardware-validated, build artifact, formal, projected, or roadmap?
- What does the AX7020 matrix prove and what does it not prove?
- What must happen before ASIC feasibility can be taken seriously?
- Which customer workload is the first proof gate?

Product Roadmap (High-Level)

Now	Zynq FPGA prototype proof paths exist. Desktop UI screenshot is hardware-validated. Workload validation is the next evidence gate.
Q3 2026	P0 workloads targeted for hardware measurement; first paid evaluation targeted; SD boot promoted only when run artifacts exist.
Q4 2026	IP diligence packet. Design partner agreement(s). ASIC feasibility study.
2027	IP licensing or strategic partnership. ASIC feasibility outcome determines path.

Source Code & Artifacts

Private GitHub repository. Access granted under NDA.

- hardware/ — ATOMiK RTL, Zynq integration, synthesis/build artifacts
- hardware/zynq/ — Zynq SoC, FSBL, NaxRiscv integration
- software/ — SDK/runtime examples and tests; cite exact test counts only after rerun
- math/proofs/ — Lean4 formal proofs of algebraic properties
- results/ — Raw measurement artifacts with evidence labels